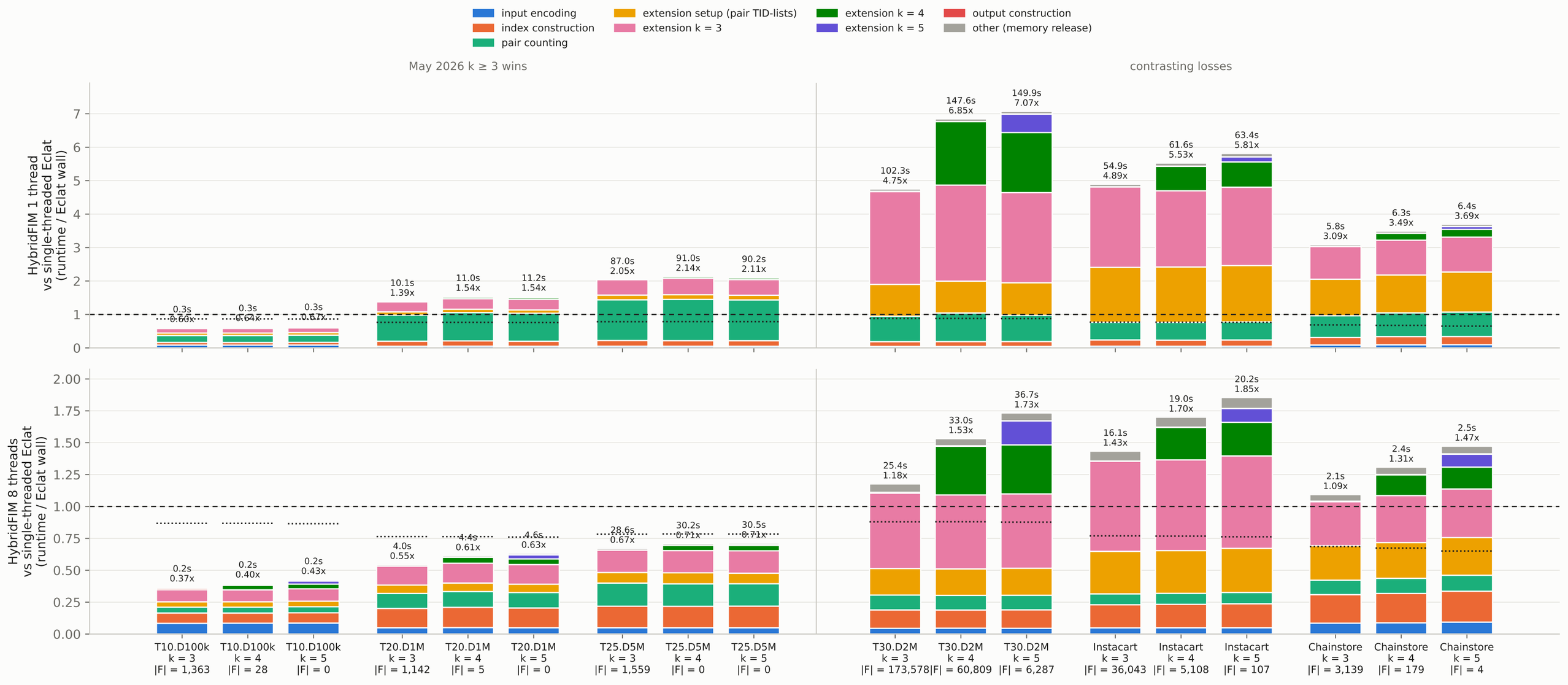
input encoding
index construction
pair counting
extension setup (pair TID-lists)
extension k = 3
extension k = 4
extension k = 5
output construction
other (memory release)
May 2026 k ≥ 3 wins
contrasting losses
HybridFIM 1 thread
vs single-threaded Eclat
(runtime / Eclat wall)
HybridFIM 8 threads
vs single-threaded Eclat
(runtime / Eclat wall)
0.3s
0.60x
0.3s
0.64x
0.3s
0.67x
10.1s
1.39x
11.0s
1.54x
11.2s
1.54x
87.0s
2.05x
91.0s
2.14x
90.2s
2.11x
102.3s
4.75x
147.6s
6.85x
149.9s
7.07x
54.9s
4.89x
61.6s
5.53x
63.4s
5.81x
5.8s
3.09x
6.3s
3.49x
6.4s
3.69x
0.2s
0.37x
0.2s
0.40x
0.2s
0.43x
4.0s
0.55x
4.4s
0.61x
4.6s
0.63x
28.6s
0.67x
30.2s
0.71x
30.5s
0.71x
25.4s
1.18x
33.0s
1.53x
36.7s
1.73x
16.1s
1.43x
19.0s
1.70x
20.2s
1.85x
2.1s
1.09x
2.4s
1.31x
2.5s
1.47x
T10.D100k k = 3 |F| = 1,363
T10.D100k k = 4 |F| = 28
T10.D100k k = 5 |F| = 0
T20.D1M k = 3 |F| = 1,142
T20.D1M k = 4 |F| = 5
T20.D1M k = 5 |F| = 0
T25.D5M k = 3 |F| = 1,559
T25.D5M k = 4 |F| = 0
T25.D5M k = 5 |F| = 0
T30.D2M k = 3 |F| = 173,578
T30.D2M k = 4 |F| = 60,809
T30.D2M k = 5 |F| = 6,287
Instacart k = 3 |F| = 36,043
Instacart k = 4 |F| = 5,108
Instacart k = 5 |F| = 107
Chainstore k = 3 |F| = 3,139
Chainstore k = 4 |F| = 179
Chainstore k = 5 |F| = 4